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Industrial Intelligence Chart 1.0 User Manual



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1. Overview

1.1 Development Objectives

The world is currently entering the era of Industry 4.0, and the most distinctive feature of this era is the use of information technology to drive industrial transformation. The most critical aspect of this digitalization is the visual management of data. Consequently, the static and dynamic visualization of data has become increasingly important in modern industrial control software.

Due to its efficiency, stability, and security, the C++ language continues to hold a significant market share in industrial control software. However, when it comes to handling data visualization, C++ appears inefficient and cumbersome; even when relevant charts are developed, they are rarely considered aesthetically pleasing or flexible.

This control was developed specifically to address the limitations of C++ in data visualization.

1.2 Software Feature Overview

Using Industrial Intelligence Chart 1.0 (hereinafter referred to as IICChart), it is very easy to implement the following features in your application, and the display quality of all charts is indistinguishable from that of Web HTML5 technology:

Display standard line charts, bar charts, scatter plots, pie charts, sunburst charts, radar charts, gauge charts, maps, and more;

Flexibly configure Cartesian, polar, radar, and map coordinate systems;

Provides out-of-the-box interactive components such as legends, visual mapping, data region zooming, tooltips, and data filtering, enabling multi-dimensional data filtering, view zooming, and detailed data display;

Enhanced support for multidimensional data. In addition to common multidimensional visualization tools such as parallel coordinates, traditional charts like scatter plots can now also accept data across multiple dimensions. Combined with the rich visual coding options provided by the visualMap component, data from different dimensions can be mapped to various visual channels such as color, size, transparency, and brightness;

IICChart is data-driven, meaning changes in the data drive changes in the

chart's presentation. Consequently, implementing dynamic data becomes exceptionally simple: simply retrieve and load the data, and IIChart will identify the differences between the two data sets and use appropriate animations to illustrate the changes. When combined with the `timeline` component, it enables the presentation of data information across higher temporal dimensions;

Flexible chart event handling and a rich set of callback interfaces make interaction between charts and applications more convenient;

In future versions, we will introduce more diverse and practical new features.

2. Installation and Usage

2.1 Hardware Requirements

The computer hardware required to run the IICChart 1.0 control must meet at least the following specifications:

Computer Hardware	Parameters
CPU	Dual-core processor with a clock speed of 1.6 GHz or higher
Memory	1 GB or more

2.2 Installation and Usage

2.2.1 System Requirements

The IICChart 1.0 control requires the Microsoft WebView2 Runtime component to be installed on Windows 7, Windows 8.x, and Windows 10 systems. Since Windows 11 already has WebView2 fully integrated, IICChart no longer requires the WebView2 Runtime to be installed on Windows 11.

The WebView2 Runtime can be downloaded from the Microsoft website at:

<https://developer.microsoft.com/zh-cn/microsoft-edge/webview2/>

After opening this website, locate the section shown in Figure 2-1, then click the buttons indicated by the red boxes labeled 1 and 2 in Figure 2-1 to download and install the software.



Figure 2-1: WebView2 Runtime Installer Download Page

Note:

The download in red box 1 in Figure 2-1 is the installer. Once run, this installer will automatically download and install the appropriate WebView2 Runtime based on your current system;

The file downloaded from red box 2 in Figure 2-1 allows users to select the appropriate full WebView2 Runtime package based on their system type for manual installation;

Note: After installing the WebView2 runtime using the methods above, the operating system will automatically update WebView2 as long as the system has Internet access. If the installed WebView2 Runtime version is too low when running IICChart, IICChart will display a corresponding prompt. If the system is not connected to the Internet at that time, please download the Runtime package using the method indicated by red box 2 in Figure 2-1, then copy this Runtime to the target system for installation.

2.2.2 Deployment to the Development Environment

2.2.2.1 Setting Up the Visual Studio Development Environment

After extracting the IICChart release package, follow these steps to import IICChart into your project:

Step 1: Add the Interface directory to your project directory, as shown in Figure 2-2;

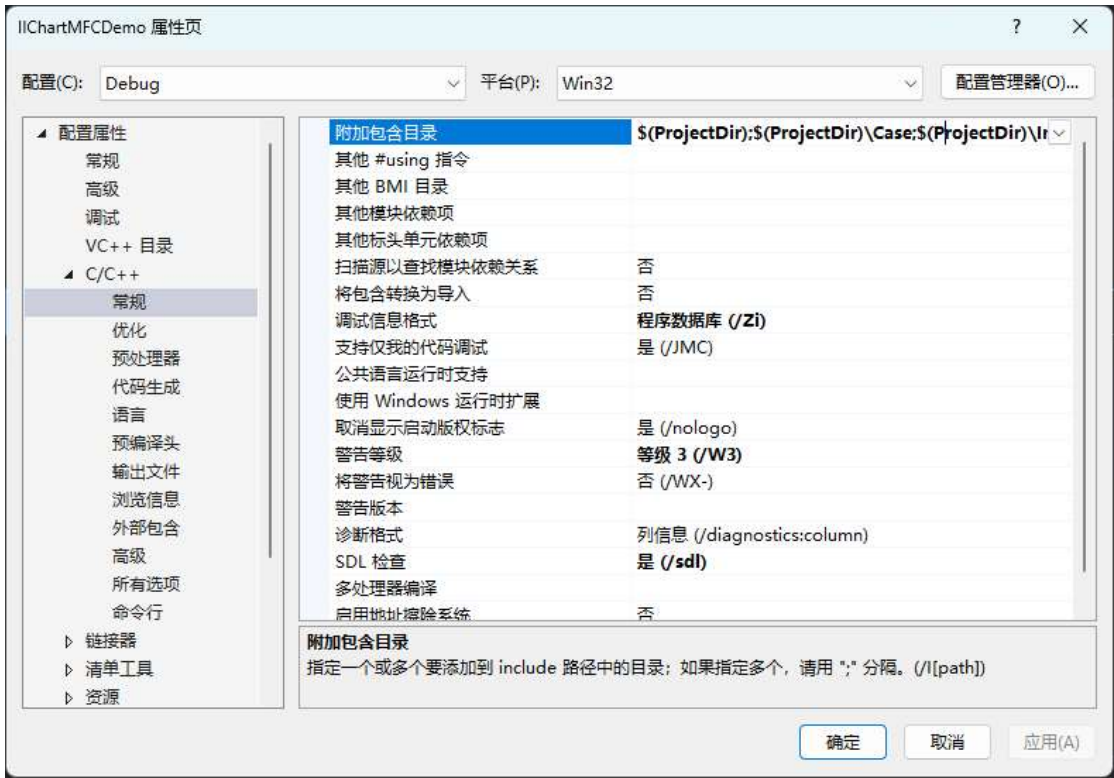


Figure 2-2: Adding the Interface directory to the project directory

- Step 2: Use the `#include` directive to reference the header file `IICChart.h` in your project;
- Step 3: Import `IICChart.dll` from the bin directory in the release package into your project;
- This completes the deployment of the IICChart VS development environment.

2.2.2.2 Setting Up the Qt Development Environment

- Step 1: Include the Interface directory in your project directory and add all referenced header files to the ".pro" file;
- Step 2: Use the “#include” directive to include the `IICChart.h` header file in your project;
- Step 3: Import `IICChart.dll` from the bin directory in the release package into your project;
- This completes the deployment of the IICChart Qt development environment.

2.2.3 Software Release

To release an application that uses IICChart, simply copy all files from the bin directory in the IICChart release package to the root directory of your application.

Note: Since IICChart relies on the WebView2 runtime, the WebView2 runtime must be installed on the end user's PC. For more information on this, please refer to Section

2.2.1.

3. Architecture

3.1 Explanation of Interface Directory Structure

After extracting the binary directory of this control, you will see two folders: bin and interface. Specifically:

The `bin` folder contains IICChart's library files and related resource files. The library files are available in both 64-bit and 32-bit versions. `IICChart.dll` is provided for dynamic loading.

The directory structure for the Interface files is as follows:

- |-- Interface: Interface directory
- |-- ActionArgs: Chart types supported by IICChart;
- |-- Callback: Callback events supported by IICChart;
- |-- Common: Includes error codes, APIs, and interfaces used for configuration;
- |-- Events: Defines IICChart event types and the parameter interfaces for event responses;
- |-- Options: Configuration options; currently supports only general configuration options;
- |--Common: Definitions of the common interfaces used by configuration options;
- |--GeneralOptions: Interface definitions for general configuration options;
- |--AngleAxis: Angular coordinate component;
- |--Axis: Coordinate axes in the Cartesian grid;
- |--DataSet: Data set component;
- |--DataZoom: Area zoom component;
- |--Geo: Geographic coordinate system component;
- |--Grid: A grid for plotting within a Cartesian coordinate system;
- |--Legend: Legend component;
- |--Polar: Polar coordinate system component;
- |--Radar: Radar chart coordinate system component;
- |--RadiusAxis: Radial axis for the polar coordinate system;
- |--Title: Title component, containing the main title and subtitle.
- |--Toolbox: Toolbar;
- |--ToolTip: Tooltip component;
- |--VisualMap: Visual mapping component;
- |--Timeline: Timeline component;
- |--StateAnimation: Component for configuring state transition animations;
- |--Series: Chart series interface definition;
- |--Bar: Bar chart;



|--Gauge: Gauge;
 |--Line: Line/area chart;
 |--Map: Map;
 |--Pie: Pie chart;
 |--Radar: Radar chart;
 |--Scatter: Scatter (Bubble) Chart;
 |--Sunburst: Sunburst chart, consisting of multiple concentric rings;

3.2 Hierarchy of interfaces

The IChart interface structure is shown in Figure 3-1. You can create an IChartManager object using the CreateChartManager() interface in IICChart.h.

Note: IChart is not thread-safe; the interfaces shown in Figure 3-1 must not be used in different threads. Unless otherwise specified, all IChart functions must be executed on the process' s UI thread.

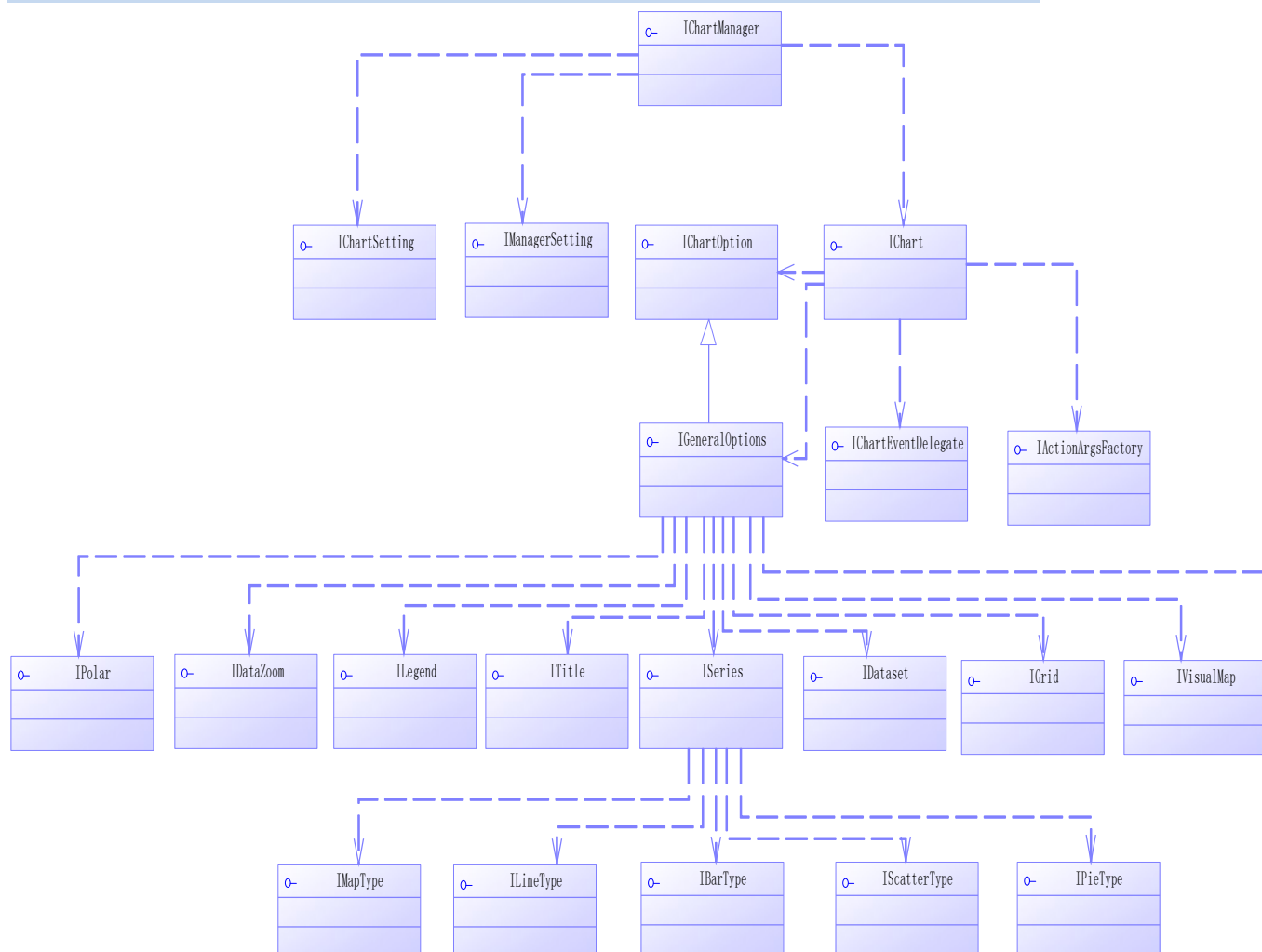


Figure 3-1 IChart Interface Layer Relationship Diagram

Multiple IChartManager objects can be created within an application, with each IChartManager corresponding to a single control in a form. This control can be any control capable of obtaining an HWND handle, such as an Edit control, a Static control, and so on.

An 'IChartManager' can create up to 20 'IChart' charts within a single viewport. Each 'IChart' has its own events to listen for and can set its own height and width. For details, refer to the API documentation.

The IChart's GetGeneralOptions() function retrieves its internal IGeneralOptions object. The IGeneralOptions object can also be used by different IChart instances. Through the IGeneralOptions object, you can define components related to the chart. For example:

- ◎ Title component: ITitle;
- ◎ Legend component: ILegend;
- ◎ Global Tooltip component: ITooltip;
- ◎ Grid component: IGrid;
- ◎ Cartesian coordinate system X-axis and Y-axis: IAxis;
- ◎ Data Zoom Component: IDataZoom;
- ◎ Polar coordinate system component: IPolar;
- ◎ Radar coordinate component: IRadar;
- ◎ Angle axis of the polar coordinate system: IAngleAxis;
- ◎ Radial axis of the polar coordinate system: IRadiusAxis;
- ◎ Geographical Coordinate System Component: IGeo;
- ◎ Series of various graphics: ISeries;
- ◎ Visual Map Component: IVisualMap;
- ◎ Dataset component: Dataset;
- ◎ Timeline component: Timeline;
- ◎ Toolbar component: Toolbox;
- ◎ State transition animation configuration component: StateAnimation;

Among these, the ISeries graphic series can create a variety of rich graphical display effects. IChart version 1.0 supports the following graphic series:

ILineType: By configuring its properties, it can display as a line chart, curve chart, area chart, step line chart, stacked line chart, or stacked area chart. When used in conjunction with IVisualMap, it can also create stacked area charts with gradient colors;



Figure 3-2 Example of a line chart

IBarType: Can be displayed as a bar chart; when used in conjunction with the polar coordinate component, it can produce even more sophisticated effects.

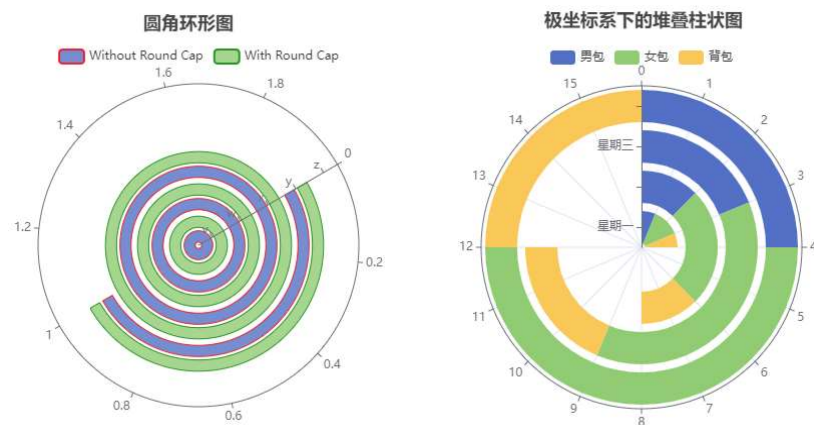


Figure 3-3: Example of a bar chart in polar coordinates

IPieType: Pie chart; through its properties, it can also be displayed as a doughnut chart, a Nightingale rose chart, and more, as shown in Figure 3-4.

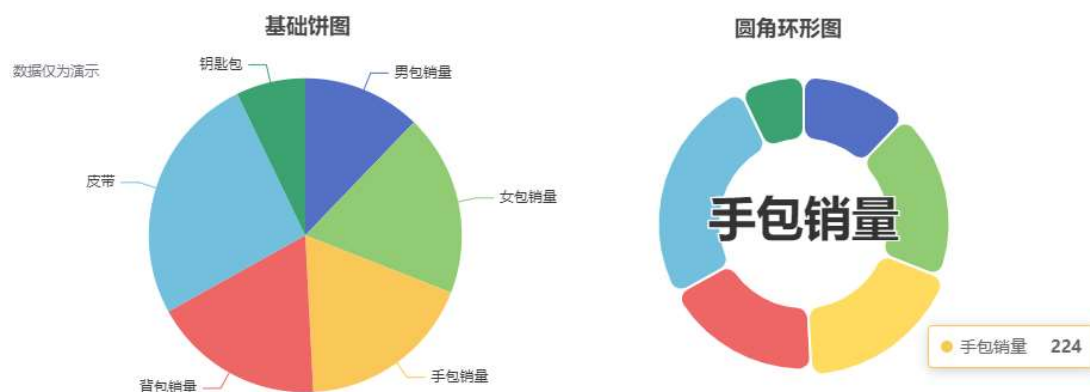


Figure 3-4: Example of a pie chart

IScatterType: Scatter plot; through its properties, it can also be displayed as a bubble chart, etc., as shown in Figure 3-5;

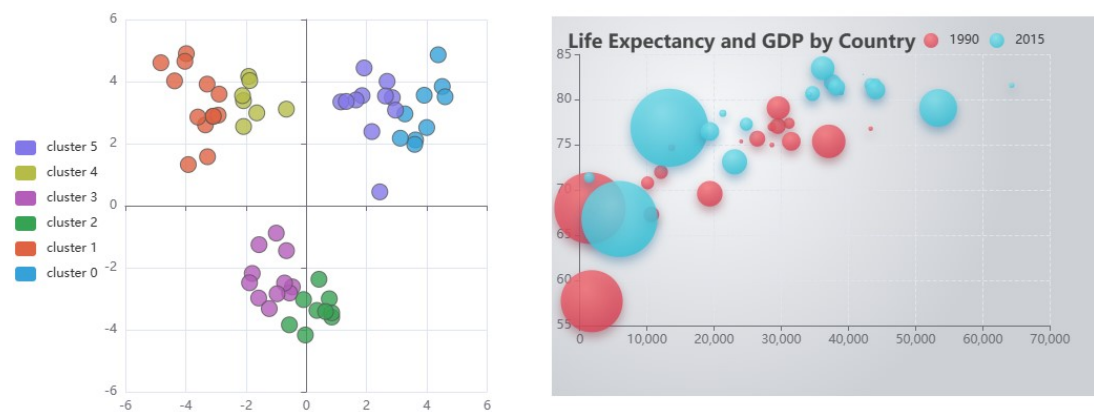


Figure 3-5 Example of a scatter plot

IRadarType: Radar chart, as shown in Figure 3-6:

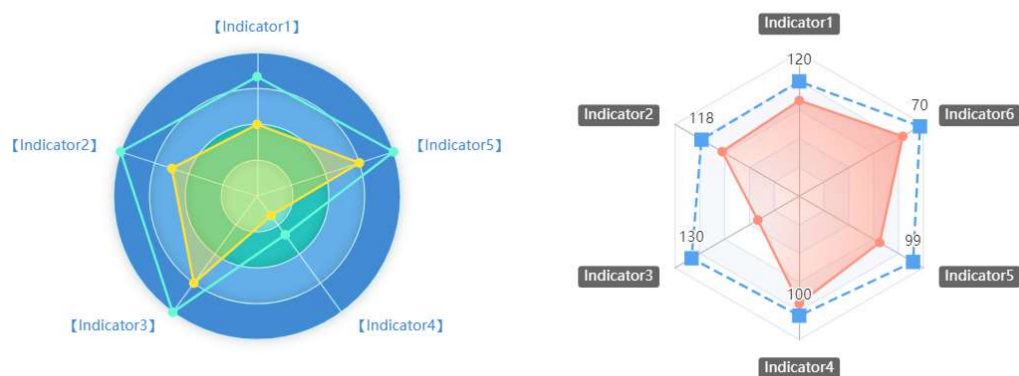


Figure 3-6 Example of a radar chart

ISunburstType: Sunburst chart, as shown in Figure 3-7:

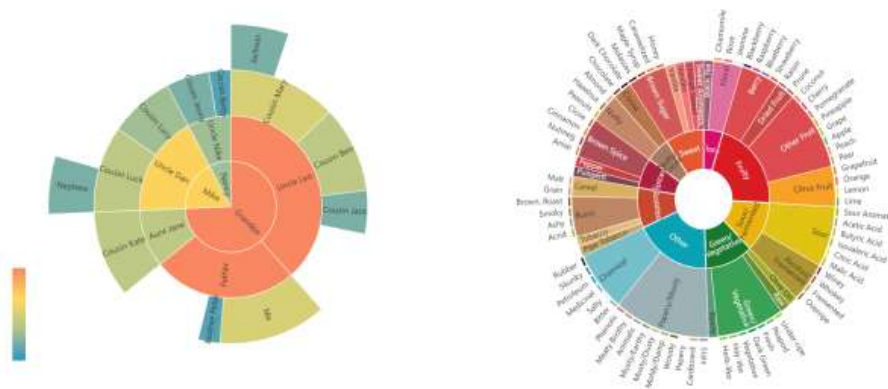


Figure 3-7: Example of a Sunburst Chart

IGaugeType: Gauge chart, as shown in Figure 3-8:

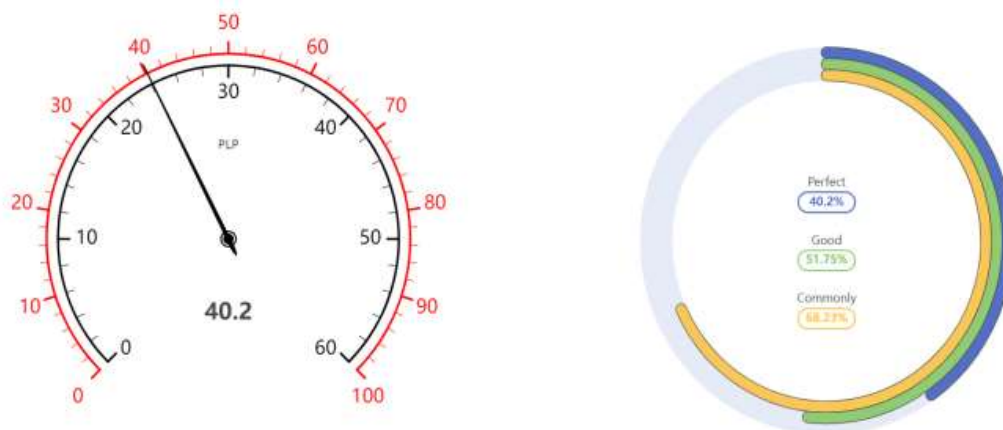


Figure 3-8: Example of a gauge chart

IMapType: Map. Supports importing map data in GeoJSON or SVG formats to display rich map effects, such as a map of China, a shopping mall wayfinding map, or a production line workstation layout diagram. As shown in Figure 3-9:

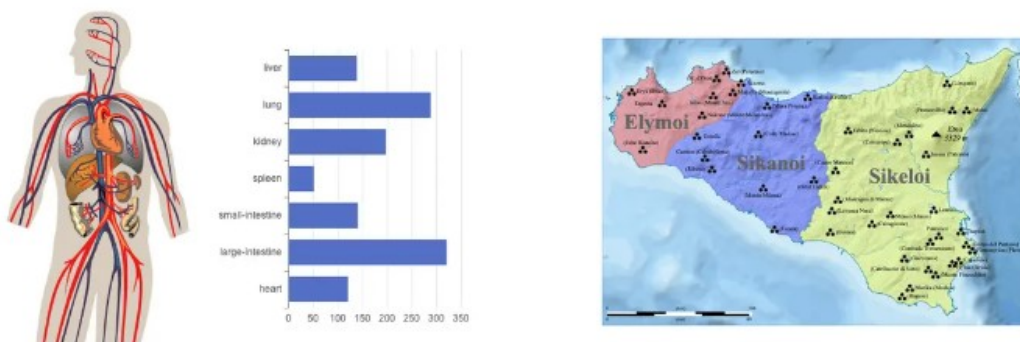


Figure 3-9: Map Example

3.3 Thread Model

IICart 1.0 is implemented using Microsoft WebView2 technology at the core. WebView2 is a secondary wrapper around Google Chrome's CEF. Therefore, IICart shares the same process model as CEF. Consequently, applications integrating IICart will run multiple processes at runtime. Process management is handled by WebView2, so you do not need to concern yourself with process resource management.

For more details on the CEF process model, please refer to the relevant technical documentation for Google Chrome or Microsoft WebView2.



4. API Documentation

4.1 ActionArgs

Chart behaviors supported by the IChart object are triggered via IChart->DispatchAction. The names and descriptions of each behavior interface are shown in the table below:

Action Interface	Function Description
IDataZoomTakeGlobalCursorAction	Toggles the selected state of the dataZoom in the toolbox.
IDownplayAction	Removes the highlight from the specified data graphic.
IGeoDownplayAction	Removes the highlight from the specified data feature in the map component.
IGeoHighlightAction	Highlights the specified data feature in the map component.
IGeoSelectAction	Selects the specified map area in the map component.
IGeoToggleSelectAction	Toggles the selection status of the specified map area.
IGeoUnSelectAction	Deselects the specified map area.
IHideTipAction	Hide the tooltip.
IHighlightAction	Highlight the specified data feature.
ILegendAllSelectAction	Select all legend entries.
ILegendInverseSelectAction	Deselect all legend entries.
ILegendScrollAction	Controls the scrolling of the legend.
ILegendSelectAction	Select the legend.
ILegendToggleSelectAction	Toggle the selection of the legend.
ILegendUnSelectAction	Deselect the legend.
ISelectAction	Select the specified data. The selected data will use the style configured in 'select'.
ISelectDataRangeAction	Selects a range of values for the map in VisualMap.
IShowTipAction	Displays a tooltip.
ITimelineChangeAction	Sets the current time point.
ITimelinePlayChangeAction	Toggle the timeline's playback status.
IToggleSelectedAction	Toggle the selected state.
IToolboxRestoreAction	Reset options.
IUnselectAction	Deselect the specified data.
IAction	Base class.
IBrushAction	This action is triggered while a brush selection is in progress; it can set or clear the selection checkbox in the chart.
IBrushEndAction	This action is automatically triggered when the brush action is complete; its parameters are exactly the same as those of the



	brush action.
IBrushTakeGlobalCursorAction	Toggle for brush mode. Use this action to set the current mouse cursor to brush mode.
ICommonAction	General action.
ICommonGeoAction	Actions related to the map component; the map component must be imported before these can be used.
IDataZoomAction	Data area zoom.

4.2 Common

Defines global enumeration types, error codes, and common object interfaces, where the common objects IFixedQueue and IQueue are created by IGlobalUtility. The interface names and brief descriptions are as follows:

Interface Name	Function Description
IEnum	Public enumeration type.
IFixedQueue	A fixed-length queue, primarily used in dynamic charting scenarios.
IQueue	A variable-length queue, primarily used when data consists of multiple types.
ErrorCode	Error code.
ICallback	Callback interface.

4.3 Events

This defines the event parameters corresponding to all chart events. In IChart, events to be listened to are primarily registered via the On interface, and the corresponding events are ultimately handled by instantiating the IChartEventDelegate interface. All events supported in version 1.0 are defined in the IEvents class. The following are the parameters for all current events:

Event Parameter Name	Parameter Description
IDataZoomEventArgs	Event arguments for the data region after zooming.
IDownplayEventArgs	Event arguments for the de-highlight event.
IDataViewChangedEventArgs	Event arguments for changes to the data view in the toolbar.
IGeoDownplayEventArgs	Event arguments for the geo de-highlight event in the geographic coordinate system.
IGeoHighlightEventArgs	Event arguments for the geo highlight event of a geographic coordinate system.
IGeoRoamEventArgs	Geo coordinate system zoom and pan event arguments.

IGeoSelectedEventArgs	Event arguments for when a map area is selected in GEO.
IGeoUnselectedEventArgs	Event arguments for when a map area in geo is deselected.
IGlobalCursorTakenEventArgs	Event arguments for the roaming mode.
IGraphRoamEventArgs	Event arguments for the zoom and pan events of a graph series.
IHighlightEventArgs	Highlight event arguments.
ILegendInverseSelectEventArgs	Event arguments for legend inverse selection.
ILegendScrollEventArgs	Legend scroll event arguments.
ILegendSelectAllEventArgs	Event arguments for the legend after selecting all items.
ILegendSelectChangedEventArgs	Event arguments for the legend selection state change.
ILegendSelectedEventArgs	Event arguments after the legend is selected.
ILegendUnselectedEventArgs	Event arguments after the legend is deselected.
IMagicTypeChangedEventArgs	Event arguments for the dynamic type switching event in the toolbar.
IMouseEventArgs	Mouse event arguments.
IRestoreEventArgs	Reset the option event arguments.
ISelectChangedEventArgs	Event arguments triggered when the data selection state changes.
ITimeLineChangedEventArgs	Event arguments when the time point on the timeline changes.
ITimeLinePlayChangedEventArgs	Event arguments for an event triggered when the playback state on the timeline changes.
ITreeRoamEventArgs	Tree chart series - event arguments for the tree's zoom and pan events.
IAxisAreaSelectedEventArgs	Event arguments for the parallel axis range selection event.
IBurshEndEventArgs	Event arguments for the brush addition complete event.
IBurshEventArgs	Event arguments for brush being added.
IBurshSelectedEventArgs	Notifies the outside what parameter is currently selected.
ICommonEventArgs	General event argument.
ICommonGeoEventArgs	General event argument for the map component.
IDataRangeSelectedEventArgs	Event arguments for events triggered when the range value changes in the visual mapping component.

4.4 IActionFactory

Chart action factory class used to create all actions sent to the chart via `IChart::DispatchAction`, enabling the chart to implement rich and dynamic animation effects.

4.5 IChart

Interfaces related to chart instances. Primary functions include setting chart configuration, updating graphic settings, adjusting chart dimensions, triggering chart



behaviors, and registering chart events. For details, refer to the IChart.h file.

4.6 IChartEventDelegate

Chart event handling interface. Handles events registered via IChart::On. For details, refer to the IChartEventDelegate.h file.

4.7 IChartManager

One IChartManager object corresponds to a Charts container control in a form. Any control capable of obtaining an HWND—such as an Edit control, a Static control, or even a Button control—can serve as the Charts container.

Up to 20 charts can be created within a single IChartManager, as shown in Figure 4-1.

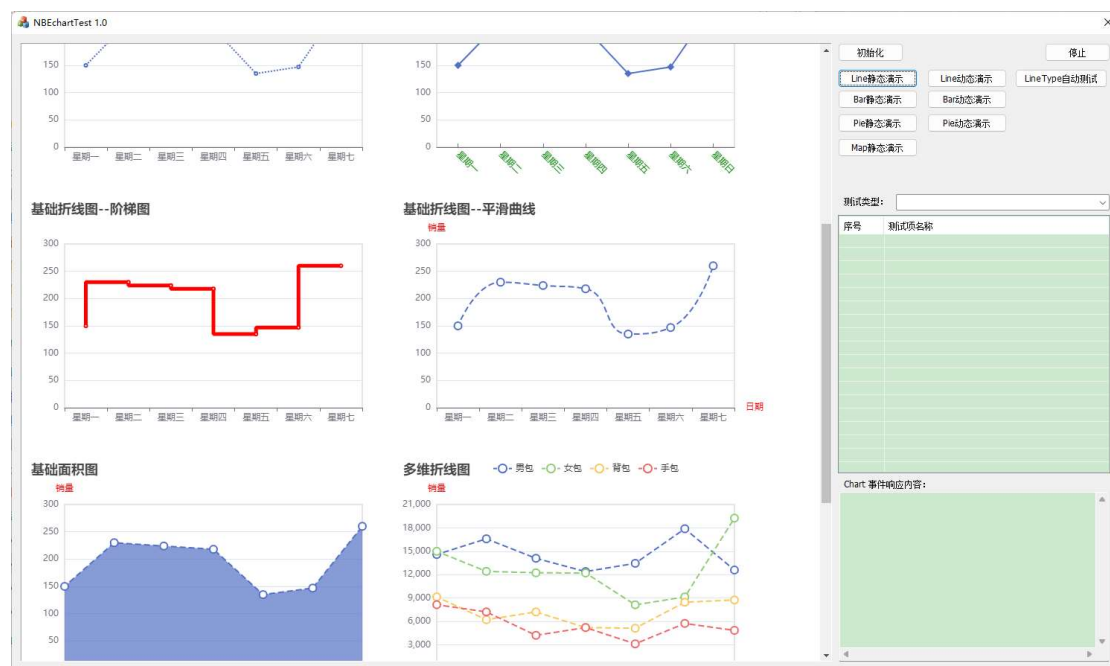


Figure 4-1: Multiple IChart objects created simultaneously in a Charts container

4.8 IChartManagerDelegate

The delegate interface for events related to the chart management class. In Version 1.0, the primary function of this interface is to allow users to handle the following events:

- ◎ An event triggered when the ChartManager initialization is complete;
- ◎ An event triggered when the view content is printed in PDF format;
- ◎ An event triggered when the container view's content is saved as an

image;

If you need to handle these events, simply instantiate this class. For details, refer to the documentation in the IChartEventDelegate.h file.

4.9 IManagerSetting

As a configuration for IChartManager, three properties are currently provided to set the Flex layout of each subchart within the chart container:

FlexDirection: The arrangement direction of each subchart;

FlexWrap: How subcharts wrap to the next line when they do not fit on a single axis;

JustifyContent: The alignment of each subchart along the main axis;

The specific interface definition is relatively simple; please refer to the IManagerSetting file definition.

4.10 IGlobalUtility

Interface for creating global public objects. Examples include creating IFixedQueue objects, IQueue objects, and IColor objects. For details, refer to the IGlobalUtility.h file.

4.11 IChartSetting

The primary function of Version 1.0 is to configure chart dimensions, language, rendering mode, theme styles, and more. For details, refer to the IChartSetting.h file.

4.12 IICChart

Creates the chart manager and global application interface, and includes all header files. The main interfaces are as follows:

- ◎ CreateChartManager: Creates an IChartManager object;
- ◎ CreateGlobalUtility: Creates an IGlobalUtility object.

5. Configuration Interface Usage Guide

General configuration options supported by the current version of the chart. The configuration of the current chart can be updated via the IChart SetOption interface. All modifications to the chart's parameters and data can be performed through

SetOption; IICChart automatically merges the new parameters and data and then refreshes the chart. If animation is enabled, IICChart identifies the differences between two sets of data and uses appropriate animations to visualize the changes.

5.1 IGeneralOptions

General configuration options consist of two main parts: global components and chart series;

The global components supported in the current V1.0 version include: AngleAxis, XAxis, YAxis, DataSet, DataZoom, Geo, Grid, Legend, Polar, Radar, RadiusAxis, Timeline, Title, Toolbox, Tooltip, and VisualMap;

The current V1.0 version supports the following chart types: Bar, Gauge, Line, Map, Pie, Radar, Scatter, and Sunburst;

Detailed descriptions of the specific functions for the interfaces of the above components and chart types are provided in the corresponding header files. This section primarily introduces the functionality of each component and chart type.

5.1.1 IAngleAxis

The angle axis for polar coordinates. Its subinterfaces and functions are as follows:

Interface Name	Function Overview
IAngleAxisAxisPointer	Configuration options for the coordinate axis pointer.
IAngleAxisAxisPointerLabel	Settings related to the axis pointer label.
IAngleAxisAxisPointerLineStyle	Axis line style
IAngleAxisAxisTick	Settings related to axis tick marks
IAngleAxisMinorSplitLine	Minor split lines in the grid area of the axis; minor split lines align with minor tick marks.
IAngleAxisMinorTick	Settings related to axis minor tick marks. Note: Minor tick lines cannot be used on category axes (type: 'category').
IAngleAxisSplitArea	The split area of the axis within the grid region; not displayed by default.
IAngleAxisSplitAreaStyle	Style settings for the split area.
IAngleAxisSplitLine	The dividing line of the coordinate axes within the grid area.
IAngleAxisSplitLineStyle	Split line style.
IAngleAxis	The angle axis of the polar coordinate system.
IAngleAxisAxisLabel	Settings related to axis label text.
IAngleAxisAxisLine	Settings related to the axis line.



5.1.2 IAxis

The x/y axes in a Cartesian coordinate grid. Generally, a single grid component can only contain two x/y axes (top and bottom). If more than two x/y axes are required, the `offset` property must be configured to prevent overlapping of multiple x/y axes at the same position.

Interface Name	Function Description
IMinorSplitLine	A minor dividing line within the grid area. Minor dividing lines align with minor tick marks.
IMinorTick	Settings related to minor tick marks on the axis.
ISplitArea	A split area within the grid region of the axis; not displayed by default.
ISplitAreaStyle	Style settings for the split area.
ISplitLine	The dividing line of the axis within the grid area.
IAxis	The x/y axes in the Cartesian coordinate grid. Generally, a single grid component can only contain a maximum of two x/y axes (top and bottom). If more than two x/y axes are required, the `offset` property must be configured to prevent overlapping of multiple x/y axes at the same position.
IAxisLabel	Settings related to axis tick labels.
IAxisLine	Settings related to axis lines.
IAxisLineStyle	Axis line style.
IAxisNameTextStyle	Text style for the axis name.
IAxisPointer	Axis pointer configuration options.
IAxisPointerLabel	Settings related to the axis pointer label.
IAxisTick	Settings related to axis ticks.

5.1.3 IDataset

A dataset component used for declaring individual datasets, allowing data to be managed separately, reused by multiple components, and freely mapped from data to visual elements. This provides significant convenience in many scenarios.

Although data can be configured for each series in the Data section of the chart series, IICart strongly recommends using datasets to manage data. This allows data to be reused by multiple components and facilitates a configuration style that separates “data” from “other settings.” After all, data is what changes most frequently at runtime, while other settings typically remain constant.

Storing data in a dataset offers the following benefits:



◎ It aligns with common data visualization workflows: (I) provide data, (II) specify the mapping from data to visuals, thereby creating a chart.

◎ Data and other configurations can be separated. Data frequently changes, while other configurations typically remain constant. Separating them facilitates independent management.

◎ Data can be reused across multiple series or components; in scenarios involving large datasets, there is no need to create a separate dataset for each series.

◎ Supports more common data formats, such as two-dimensional arrays and object arrays, thereby reducing the need for users to convert data formats.

The following is an overview of the sub-interface functions of the dataset component:

Interface Name	Function Overview
IDatasetSource	Data Source: Data object or array
IDatasetTransformConfig	Configuration for the dataset component's data transformation functionality.
IDatasetTransformConfigItem	Configuration item for the Dataset component's data transformation functionality.
IDataset	A dataset allows data to be managed separately and reused by multiple components, and allows for flexible mapping of data to visualizations. This offers significant convenience in many scenarios, specifically: 1. It aligns with common data visualization paradigms: (I) Provide data, (II) Specify the mapping from data to visualizations to generate charts. 2. Data and other configurations can be separated. Data frequently changes, while other configurations typically remain constant. Separating them facilitates independent management. 3. Data can be reused across multiple series or components. In scenarios involving large datasets, there is no need to create a separate dataset for each series. 4. Supports common data formats, such as two-dimensional arrays and arrays of objects. This helps users avoid having to convert data formats to a certain extent.
IDatasetDimensionsData	Each item in the dimensions array; It is worth noting that when dimensions are defined, the default tooltip display for each dimension will switch to a "vertical layout," making it easier to display the name of each dimension. If dimensions are not defined, the default tooltip displays values horizontally and shows only the numerical values without dimension names.

	Use 'dimensions' to define the information for each dimension of 'series.data' or 'dataset.source'. Note: If using a dataset, you can define dimensions in 'dataset.dimensions' or specify dimension names in the first row/column of 'dataset.source'. In that case, you do not need to specify dimensions here. However, if dimensions are specified here, they take precedence.
--	--

5.1.4 IDataZoom

5.1.4.1 Feature Overview

The DataZoom component is used for zooming in on specific areas, allowing you to focus on detailed data, view the data as a whole, or filter out outliers.

The following types of DataZoom components are currently supported:

- ◎ Embedded DataZoom component (DataZoomInside): Embedded within a coordinate system, allowing users to zoom or pan the coordinate system by dragging with the mouse, using the mouse wheel, or swiping with a finger (on a touchscreen).
- ◎ DataZoomSlider: Features a dedicated slider that users can use to zoom or pan.
- ◎ Selection-based Data Zoom Component (DataZoomSelect): Provides a selection box for zooming the data area. That is, IToolbox→IFeature→IDataZoom.

The effect of the slider-type data area zoom component is shown in Figure 5-1:

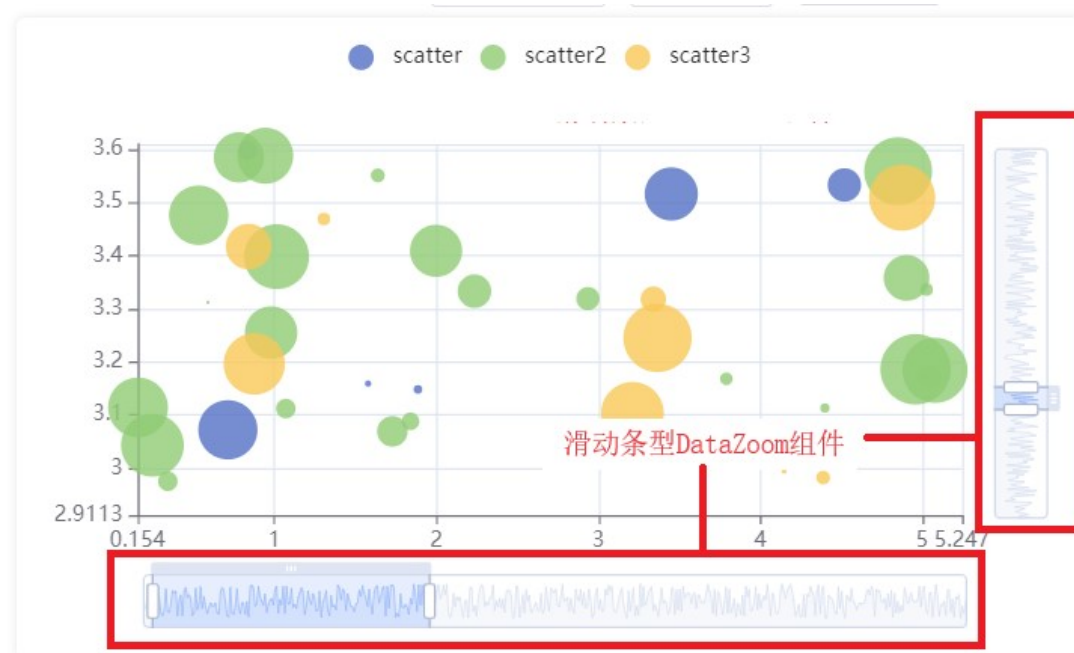


Figure 5-1 Slider-type DataZoom component

5.1.4.2 The Relationship Between DataZoom and the Axis

DataZoom primarily operates on axes (controlling the display range of the axes, also known as the window).

You can specify which axis or axes DataZoom controls using `xAxisIndex`, `yAxisIndex`, `radiusAxisIndex`, or `angleAxisIndex`.

Multiple DataZoom components can exist simultaneously, working together to control the axis. If multiple DataZoom components jointly control the same axis, they will automatically synchronize.

5.1.4.3 How the DataZoom Component Affects Axes and Data

DataZoom achieves the effect of data window zooming by filtering data and internally setting the display window of the axis.

Different settings for the data filtering mode (`dataZoom.filterMode`) produce different results. Available values are:

EFM_Filter: Data outside the current data window is filtered out. This affects the data range of other axes. For each data item, if even one dimension falls outside the data window, the entire data item is filtered out.

EFM_WeakFilter: Data outside the current data window is filtered out. This affects the data range of other axes. For each data item, the entire item is filtered out only if all dimensions are outside the data window on the same side.

EFM_Empty: Data outside the current data window is set to empty. This does not affect the data range of other axes.

EFM_None: Does not filter data; only changes the axis range.

The user should decide how to configure this based on the specific scenario and requirements. As a general rule:

- ◎ When "only the X-axis or only the Y-axis is controlled by the dataZoom component," `filterMode: EFM_Filter` is commonly used. This allows the other axis to adapt to the filtered data range.
- ◎ When the X-axis and Y-axis are each controlled by a dataZoom component:
 - > If the X-axis and Y-axis are "equal in status and should not affect each other," such as in a "dual-axis scatter plot," then both axes can be set to `filterMode: EFM_Empty``.
 - > If the X-axis is primary and the Y-axis is secondary—such as in a

'bar chart'—where you need to 'drag dataZoomX to change the filtered bars on the X-axis while the Y-axis range automatically adapts to the height of the remaining bars', and "when dragging `dataZoomY` to filter bars on the Y-axis, the X-axis range remains unaffected," then set the X-axis to `filterMode: EFM\_Filter` and the Y-axis to `filterMode: EFM\_Empty`, i.e., the primary axis is `EFM\_Filter` and the secondary axis is `EFM\_Empty`.

The following is a brief introduction to the functions of the DataZoom component's sub-interfaces:

Interface Name	Function Overview
IDataZoomSliderBrushStyle	Brush style settings.
IDataZoomSliderEmphasis	Highlight style settings.
IDataZoomSliderHandleStyle	Style configuration for the zoom handles on both sides.
IDataZoomSliderMoveHandleStyle	Style configuration for the move handle.
IDataZoomSliderSelectedDataBackground	Style configuration for the shadow of the selected data section.
IDataZoomSliderTextStyle	Text style for axis labels.
IDataZoom	The dataZoom component is used for regional zooming, allowing you to focus on detailed data details, get an overview of the data as a whole, or eliminate the influence of outliers.
IDataZoomAreaStyle	Fill style for the shadow.
IDataZoomDataBackground	The style of the data shadow.
IDataZoomInside	Built-in data area zoom component, "Built-in" refers to being integrated into the coordinate system.
IDataZoomSlider	A segmented visual mapping component with three modes: Continuous data with equal segmentation: Automatically divides the data into equal segments based on `visualMap-pieewise.splitNumber`. Custom segmentation for continuous data: Defines the range of each segment based on `visualMap-pieewise.pieces`. Segmentation of discrete data by category: Categories are defined in `visualMap-pieewise.categories`.

5.1.5 IGeo

Geographic Coordinate System Component. The Geographic Coordinate System



Component is used for map rendering and supports plotting scatter plots (ScatterType) and line sets (Lines) on geographic coordinate systems. The following is a brief introduction to the features of the geographic component sub-interfaces:

Interface Name	Function Overview
IGeoLabel	A text label on a graphic that can be used to describe certain data information about the graphic, such as values, names, etc.
IGeoProjection	A custom map projection that must provide at least two methods: 'project' and 'unproject', used to calculate projected coordinates and unprojected coordinates, respectively.
IGeoRegion	Configures styles for specific regions on a map.
IGeoScaleLimit	Control the zoom limits of the scroll wheel using the minimum and maximum zoom values; the default zoom level is 1.
IGeoSelect	Polygon and label styles when selected.
IGeoTooltip	Series-specific tooltip settings.
IGeoTooltipTextStyle	Text style for the tooltip overlay.
IGeo	The geographic coordinate system component is used for map rendering and supports plotting scatter plots and line sets on the geographic coordinate system.
IGeoBlur	The style of polygons and labels in the faded state.
IGeoEmphasis	Polygon and label styles in the highlighted state.
IGeoEmphasisEx	Polygon and label styles in the highlighted state.
IGeoItemStyle	Polygon graphic styles for map areas.

5.1.6 IGrid

A grid for plotting in a Cartesian coordinate system; a single Grid can contain up to two X-axes (vertical and horizontal) and two Y-axes (left and right). Line charts, bar charts, and scatter plots (bubble charts) can be plotted on the grid. In IICart Version 1.0, any number of Grid components can be present; the effect is shown in Figure 5-2:

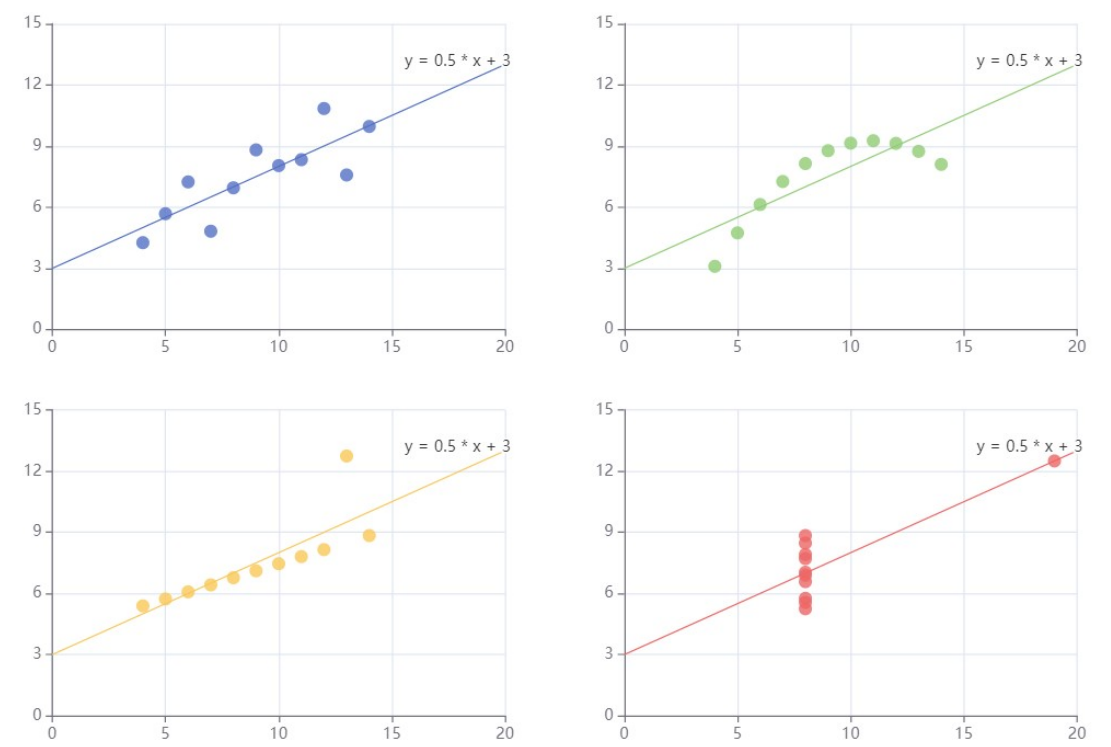


Figure 5-2 Multi-Grid Example

The following is an overview of the Grid component’ s sub-interface functions:

Interface Name	Function Overview
IGridTooltipAxisPointerLabel	Settings related to axis tick labels.
IGridTooltipAxisPointerLineStyle	Line style settings.
IGridTooltipTextStyle	Text style settings for the tooltip overlay.
IGrid	A grid for plotting within a Cartesian coordinate system; a single grid can contain up to two X-axes (vertical and horizontal) and two Y-axes (left and right). Line charts, bar charts, and scatter plots (bubble charts) can be plotted on the grid.
IGridTooltip	Tooltip settings specific to this coordinate system.
IGridTooltipAxisPointer	Configuration options for axis pointers.
IGridTooltipAxisPointerCrossStyle	Line style settings.

5.1.7 ILegend

Legend component. The legend component displays the symbols, colors, and names of different series. You can control which series are not displayed by clicking the legend and selecting " ".

A single IChart instance in IICart can contain multiple legend components, which facilitates the layout of multiple legends. When there are too many legends, you can use Scrollable Legend (Vertical) or Scrollable Legend (Horizontal).

The legend is shown in Figure 5-3:

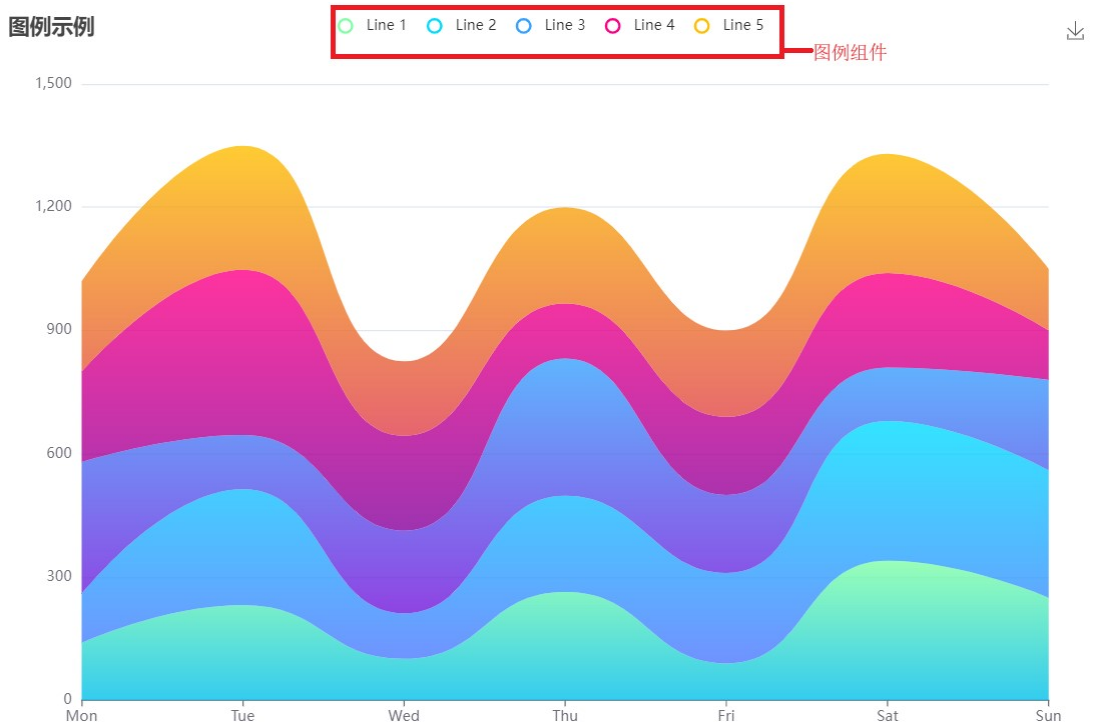


Figure 5-3 Legend Effect

The following is an overview of the Legend component’ s sub-interface features:

Interface Name	Function Overview
ILegendLineStyle	The style of the line in the legend graphic, used for setting the style of horizontal lines in line chart legends, for example.
ILegendPageIcons	Icons for the legend control block.
ILegendPageTextStyle	Text style for legend page information.
ILegendSelectorLabel	Text label style for the selector button; displayed by default.
ILegendTextStyle	The common text style for the legend.
ILegend	The legend component displays the symbols, colors, and names of different series. You can control which series are hidden by clicking the legend. A single ECharts instance can contain multiple legend components, which facilitates the layout of multiple legends.
ILegendData	Legend data.
ILegendEmphasis	Highlight style settings for the legend.
ILegendItemStyle	Graphic style of the legend.

5.1.8 IPolar

Polar coordinate system, which can be used for scatter plots and line charts. Each polar coordinate system has an angle axis and a radius axis. The result is shown in

Figure 5-4:

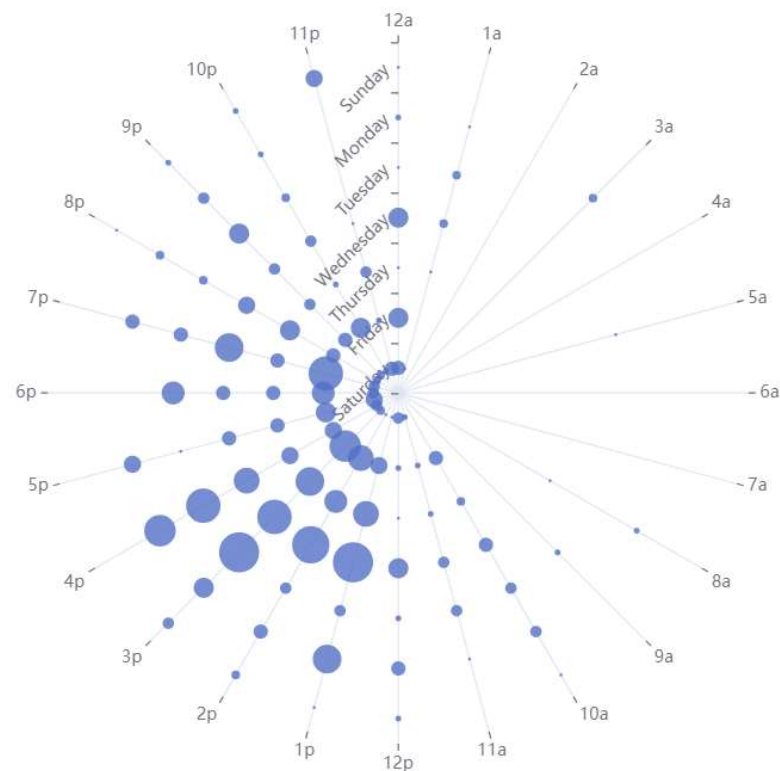


Figure 5-4 Polar Coordinate Example

The following is an overview of the Polar component’ s sub-interfaces:

Interface Name	Function Overview
IPolarTooltipAxisPointerCrossStyle	The style of the crosshairs for the axis pointer in the tooltip specific to this coordinate system.
IPolarTooltipAxisPointerLabel	The text label for the axis pointer of the tooltip specific to this coordinate system.
IPolarTooltipAxisPointerLineStyle	Line style settings.
IPolarTooltipTextStyle	The text style of the tooltip overlay.
IPolar	Polar coordinate system, suitable for scatter plots and line charts. Each polar coordinate system has an angle axis and a radius axis.
IPolarTooltip	Tooltip settings specific to this coordinate system.
IPolarTooltipAxisPointer	Configuration options for the axis pointer.

5.1.9 IRadar

Radar chart coordinate system component; radar charts use the Radar component as their coordinate system. Unlike polar coordinate systems, each axis in a radar chart is

a separate dimension. The style of the axis lines can be configured using the Name, AxisLine, AxisTick, AxisLabel, SplitLine, and SplitArea properties. The result is shown in Figure 5-5

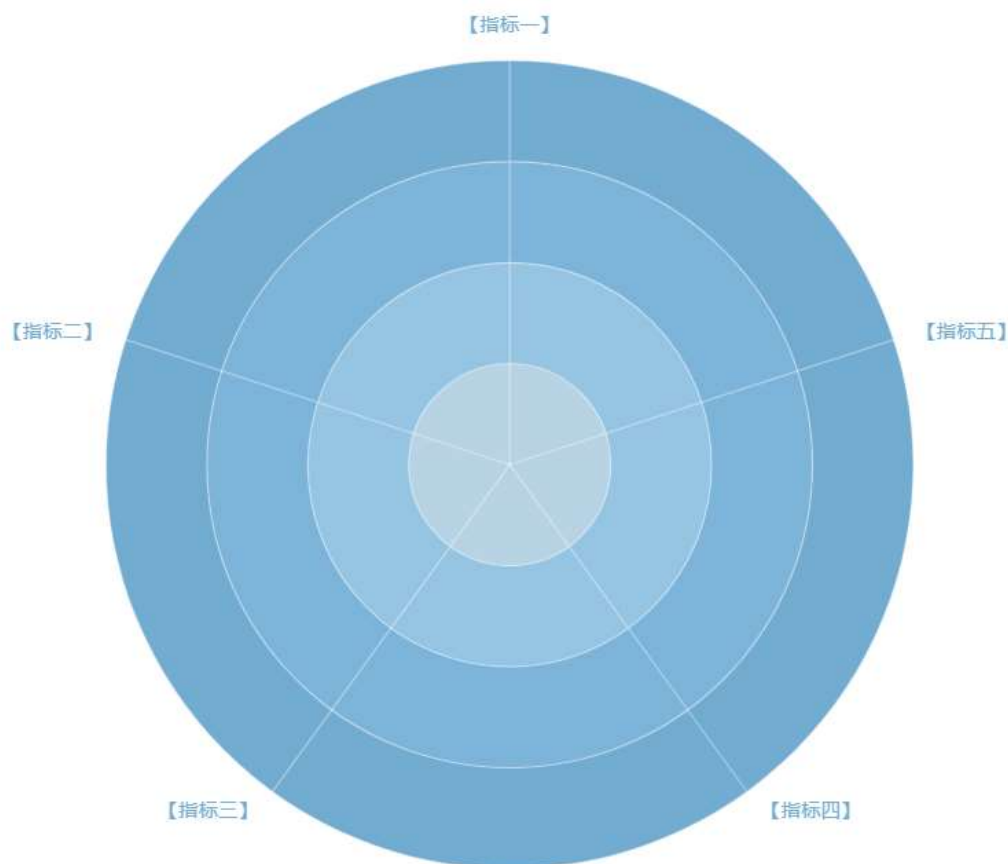


Figure 5-5 Radar Chart Example

The following is an overview of the sub-interface functions of the Radar Chart component:

Interface Name	Function Overview
IRadarIndicator	The radar chart indicator, used to specify multiple variables (dimensions) in the radar chart.
IRadarSplitArea	A split area for the coordinate axes within the grid region; not displayed by default.
IRadarSplitAreaStyle	Style settings for the split area.
IRadarSplitLine	The dividing line of the coordinate axes within the grid area.
IRadarSplitLineStyle	Split line style.
IRadar	Radar chart coordinate system component; applicable only to radar charts.
IRadarAxisLabel	Settings related to axis label text.
IRadarAxisLine	Settings related to axis lines.

IRadarAxisName	Configuration options for the name of each indicator in the radar chart.
IRadarAxisTick	Settings related to axis tick marks.

5.1.10 IRadiusAxis

A radial axis component for the polar coordinate system. The following is a brief introduction to the functions of this component's sub-interfaces:

Interface Name	Function Overview
IRadiusAxisAxisTick	Settings related to axis tick marks.
IRadiusAxisMinorSplitLine	Minor tick lines on the axis within the grid area. Minor tick lines align with minor tick marks.
IRadiusAxisMinorTick	Settings related to axis tick marks.
IRadiusAxisNameTextStyle	Text style for the axis name.
IRadiusAxisSplitArea	The split area of the axis within the grid region; not displayed by default.
IRadiusAxisSplitAreaStyle	Style settings for the split area.
IRadiusAxisSplitLine	The dividing line of the axis within the grid area.
IRadiusAxisSplitLineStyle	Split line style.
IRadiusAxis	The radial axis of the polar coordinate system.
IRadiusAxisAxisLabel	Settings related to axis label text.
IRadiusAxisAxisLine	Settings related to the axis line.
IRadiusAxisAxisPointer	Configuration options for the axis pointer.
IRadiusAxisAxisPointerLabel	Settings related to axis tick labels.

5.1.11 IStateAnimation

Configuration for state transition animations; supports setting series-specific configurations within each series. The following is a brief introduction to the functions of this component's subinterfaces:

Interface Name	Function Overview
IStateAnimation	Configuration for state transition animations, supporting series-specific settings for each series.

5.1.12 ITimeline

Timeline component, providing functionality to switch between and play multiple ECharts options. The effect is shown in Figure 5-6:





Figure 5-6 Example of the Timeline Component

The following is an overview of the sub-interfaces of the Timeline component:

Interface Name	Function Overview
ITimeLineStyle	Axis style settings.
ITimeLineProgress	Get the styles of the line, inflection points, and labels in the progress bar.
ITimeLineProgressLabel	Progress bar labels.
ITimeLineProgressLineStyle	Progress line style.
ITimeLine	Timeline component that provides functionality for switching between multiple options, playback, and other operations.
ITimeLineCheckpointStyle	Current item graphic style.
ITimeLineControlStyle	The style of the control buttons.
ITimeLineData	Data.
ITimeLineEmphasis	Highlight state.
ITimeLineEmphasisLabel	Text label.
ITimeLineItemStyle	Graphic style.
ITimeLineLabel	Text label.

5.1.13 ITitle

Title component, containing the main title and subtitle. Titles can be formatted in IChart. The following is an overview of the Title component's subinterface features:

Interface Name	Function Overview
ITitleSubTextStyle	Subtitle style.
ITitleTextStyle	Main title style.
ITitle	Title component, containing the main title and subtitle.

5.1.14 IToolbar

Toolbar, featuring five built-in tools: Export Image, Data View, Dynamic Type Switch, Data Area Zoom, and Reset. The following is an overview of the toolbar component's sub-interface functions:

Interface Name	Function Overview
IToolboxFeatureBrushTitle	Title text.
IToolboxFeatureDataView	Data view.
IToolboxFeatureDataZoom	Data area zoom.
IToolboxFeatureDataZoomBrushStyle	Zoom brush style.
IToolboxFeatureDataZoomIcon	The icon path for zooming in and out.
IToolboxFeatureDataZoomTitle	Title text for zoom and restore.
IToolboxFeatureRestore	Restore configuration settings.
IToolboxFeatureSaveAsImage	Save as Image.
IToolboxIconStyle	Common icon style settings.
IToolbox	Global toolbar with five built-in tools: Export Image, Data View, Dynamic Type Switch, Data Area Zoom, and Reset.
IToolboxEmphasis	Common icon style settings. Since the text information for icons is only displayed when the cursor hovers over them, please configure text-related options under "Emphasis".
IToolboxFeature	Configuration options for each tool. In addition to the built-in tool buttons, you can also customize tool buttons.
IToolboxFeatureBrush	Control buttons for the selection box.
IToolboxFeatureBrushIcon	The icon path for the button.

5.1.15 ITooltip

Tooltip component. The tooltip component can be configured in several places:

Globally, i.e., ITooltip

It can be set in a coordinate system, such as IGrid.Tooltip, IPolar.ITooltip, or ISingle.ITooltip

Within a series, i.e., `ISeries.ITooltip`

It can be set for each data item in a series, i.e., ISeries.IData.Tooltip

The following is a brief introduction to the functionality of the Tooltip component's subinterfaces:

Interface Name	Function Overview
ITooltipAxisPointerCrossStyle	The style of the crosshair on the tooltip's axis indicator.
ITooltipAxisPointerLabel	Settings related to axis tick labels.
ITooltipAxisPointerLineStyle	The style of the tooltip's axis pointer line.
ITooltipTextStyle	Tooltip text style.



ITooltip	Tooltip component.
ITooltipAxisPointer	Coordinate axis pointer configuration option; `tooltip.axisPointer` is a shortcut for configuring the coordinate axis pointer. In fact, all functions of the coordinate axis pointer can be implemented via the `axisPointer` configuration option on the axis: ``

5.1.16 IVisualMap

5.1.16.1 Feature Overview

VisualMap is a visual mapping component used for "visual encoding," which involves mapping data to visual elements (visual channels). Visual elements can be:

- ◎ **Symbol**: The graphical category of the symbol.
- ◎ **SymbolSize**: The size of the symbol.
- ◎ **Color**: The color of the symbol.
- ◎ **ColorAlpha**: The transparency of the symbol's color.
- ◎ **Opacity**: The opacity of the symbol and its attachments (such as text labels).
- ◎ **ColorLightness**: The lightness of the color; see HSL.
(URL: https://en.wikipedia.org/wiki/HSL_and_HSV)
- ◎ **ColorSaturation**: The saturation of the color; see HSL.
- ◎ **ColorHue**: The hue of a color; see HSL.

Multiple VisualMap components can be defined, allowing for simultaneous visual mapping of multiple dimensions in the data. VisualMap components can be defined as piecewise (VisualMapPiecewise) or continuous (VisualMapContinuous), distinguished by the Type property. For details, refer to IVisualMap.

5.1.16.2 Configuring Visual Mapping Methods

Since this is a mapping from "data" to "visual elements," you can specify in the visualMap which "data dimensions" are mapped to which "visual elements." For example, you can use `inRange` to define visual elements within a selected range. The definition format is as follows:

```
visualMap: [
  { ...,
    inRange: {
      color: ['#121122', 'rgba(3,4,5,0.4)', 'red'],
      symbolSize: [30, 100]
    }
  }
]
```



}]

The following is an overview of the features of the VisualMap component's sub-interfaces:

Interface Name	Function Overview
IVisualMapPieceComponent	An optional visual element in VisualMap; multiple instances can exist simultaneously.
IVisualMapPiecewise	A piecewise visual mapping component.
IVisualMapRange	A visual element for a specified range. (Users can interact with the visualMap component to select a range using a mouse or touch.)
IVisualMapTarget	Visual element for the target series.
IVisualMapTextStyle	Text style for the tooltip overlay.
IVisualMap	A visual mapping component used for "visual encoding," which involves mapping data to visual elements (visual channels).
IVisualMapContinuous	Continuous visual mapping component.
IVisualMapController	The inRange and outOfRange settings for the controller in the visualMap component.
IVisualMapHandleStyle	Handle style configuration.
IVisualMapIndicatorStyle	Indicator style.

5.1.17 ISeries

Manages all supported chart series.

5.1.17.1 ILineType

Line/Area Chart Series. A line chart connects data points with lines to illustrate trends in data. It can be used in both Cartesian and polar coordinate systems.

Additionally, setting AreaStyle allows you to plot area charts. When used with the segmented VisualMap component, line/area charts can be divided into sections using different colors.

The following is a brief introduction to the functions of the LineType series sub-interfaces:

Interface Name	Function Overview
ILineTypeBlur	The faded state of a line chart; effective when emphasis.focus is enabled.
ILineTypeBlurAreaStyle	Area fill style; when set, the chart is displayed as an area chart.
ILineTypeBlurEndLabel	Label for the line chart during the fade-out.
ILineTypeBlurItemStyle	Style of the inflection point markers on the line chart during the fade-out.



ILineTypeBlurLabel	Configuration for labels during the fade-out of a line chart.
ILineTypeBlurLabelLine	Configuration of the visual guide line for the line chart during fading.
ILineTypeData	An array containing the data items in the series. Array items are typically specific data entries.
ILineTypeDataBlur	Fade-out style settings for a single inflection point.
ILineTypeDataEmphasis	Style settings for highlighting a single inflection point.
ILineTypeDataItemStyle	Style settings for a single inflection point marker.
ILineTypeDataLabel	Style settings for the text of a single inflection point.
ILineTypeDataLabelLine	Configuration of visual guides for a single data item label.
ILineTypeDataSelect	Style and label settings for a single data point.
ILineTypeDecal	Decal pattern for the graphic; takes effect only when both `aria.enabled` and `aria.decal.show` are set to `true`. Takes effect only when the fill style `areaStyle` is set.
ILineTypeDimensionsData	Defines the information for each dimension of `series.data` or `dataset.source` using `dimensions`.
ILineTypeEmphasis	The highlight state of the line chart; effective when `emphasis.focus` is enabled.
ILineTypeEmphasisAreaStyle	The fill style for the highlighted area. When set, the chart displays as an area chart.
ILineTypeEmphasisEndLabel	The label displayed when a line chart is highlighted.
ILineTypeEmphasisItemStyle	The ItemStyle for the line chart when highlighted.
ILineTypeEmphasisLabel	The text label for the highlighted portion of a line chart, which can be used to provide explanatory information about the chart data.
ILineTypeEmphasisLabelLine	The visual guide line configuration for the label when the line chart is highlighted.
ILineTypeEndLabel	The label at the endpoint of a line in a line chart.
ILineTypeItemStyle	Style of the line break markers.
ILineTypeLabel	Labels on the chart that can be used to explain certain data information.
ILineTypeLabelLayout	Unified layout configuration for labels.
ILineTypeLabelLine	Configuration of the label's visual guide lines.
ILineTypeMarkArea	Chart marker area, commonly used to highlight a specific range of data in a chart, such as indicating when an advertisement was placed.
ILineTypeMarkAreaBlur	The style of the mark area during fading. The fading rules follow those of the series it belongs to.
ILineTypeMarkAreaData	The data array for the mark area.
ILineTypeMarkAreaEmphasis	Highlight style for the mark area.
ILineTypeMarkAreaItemStyle	The style of this mark area.

ILineTypeMarkAreaLabel	Text configuration for the marked area.
ILineTypeMarkLine	Chart line.
ILineTypeMarkLineBlur	Style of the chart marker line when fading out; the fading rules follow those of the series.
ILineTypeMarkLineData	Data for the start or end point of the chart marker line.
ILineTypeMarkLineDataBlur	The fade-out style for the chart marker line data.
ILineTypeMarkLineDataEmphasis	The highlighting style for chart marker line data.
ILineTypeMarkLineEmphasis	Highlight style for chart marker lines.
ILineTypeMarkLineLabel	Label style for chart marker lines.
ILineTypeMarkLineLineStyle	Line style for chart marker lines.
ILineTypeMarkPoint	Chart annotation.
ILineTypeMarkPointBlur	The fade style for the annotation; the fade rules follow those of the series it belongs to.
ILineTypeMarkPointData	Elements of the mark data array.
ILineTypeMarkPointDataEmphasis	Highlight style for the annotation data.
ILineTypeMarkPointDataItemStyle	The style of the annotation.
ILineTypeMarkPointEmphasis	Highlight style for the annotation.
ILineTypeMarkPointItemStyle	Style of the annotation data
ILineTypeSelect	The selection state of the line chart; effective when selectedMode is enabled.
ILineTypeSelectAreaStyle	Fill style for the selected area of a line chart.
ILineTypeSelectEndLabel	EndLabel for a selected data point in a line chart
ILineTypeSelectItemStyle	The style of the inflection point marker when a line chart is selected.
ILineTypeSelectLabel	Label configuration for a selected line chart.
ILineTypeSelectLabelLine	Configuration of the visual guide lines for the label when a line chart is selected.
ILineType	Line/Area Chart: A line chart connects data points with lines to illustrate trends in data. It can be used in both Cartesian and polar coordinate systems.
ILineTypeAreaStyle	Area fill style; when set, the chart is displayed as an area chart.

5.1.17.2 IBarType

A bar chart (also known as a column chart) is a common chart type that represents data magnitude through the height of bars (or width in horizontal orientation).

Enabling the RealtimeSort property enables real-time sorting functionality.

The following is a brief introduction to the sub-interfaces of the BarType chart series:

Interface Name	Function Overview
IBarTypeBlur	Graph and label styles during fade-out. Effective when



	emphasis.focus is enabled.
IBarTypeBlurItemStyle	Graphic style during fade-out.
IBarTypeBlurLabelLine	Configuration of visual guides for labels during fade-out.
IBarTypeData	An array of data content in the series; array items are typically specific data entries.
IBarTypeDataBlur	Configuration for the fade-out state of a single data item.
IBarTypeDataEmphasis	Configuration for the highlight state of a single data item.
IBarTypeDataItemStyle	Graphic style settings for a single data item.
IBarTypeDataLabel	Labels on the chart that can be used to provide information about the data, such as values, names, etc.
IBarTypeDataLabelLine	Configuration of the label's visual guide lines.
IBarTypeDataSelect	Configuration of the selection state for individual data points.
IBarTypeDecal	Decal pattern for the graphic; takes effect only when both 'aria.enabled' and 'aria.decal.show' are set to 'true', and requires that the 'areaStyle' fill style be configured.
IBarTypeDimensionsData	Defines the information for each dimension of 'series.data' or 'dataset.source' using 'dimension'.
IBarTypeEmphasis	Highlight state; takes effect when 'emphasis.focus' is enabled.
IBarTypeEmphasisItemStyle	Graph style when highlighted.
IBarTypeEmphasisLabelLine	Configuration of the visual guide lines for the label when highlighted.
IBarTypeItemStyle	Graphic style.
IBarTypeLabel	Text labels on the chart, which can be used to provide information about the chart data, such as values, names, etc.
IBarTypeLabelLayout	Unified layout configuration for labels.
IBarTypeLabelLine	Configuration of visual guides for the label.
IBarTypeMarkArea	Chart marker area, commonly used to highlight a specific range of data in a chart, such as indicating when an advertisement was placed.
IBarTypeMarkAreaBlur	Fade-out style for the marker; the fade-out rules follow those of the series it belongs to.
IBarTypeMarkAreaData	The data array elements for the marked area.
IBarTypeMarkAreaEmphasis	Highlight style for the marked area.
IBarTypeMarkAreaItemStyle	The style of this mark area.
IBarTypeMarkAreaLabel	Text configuration for the marked area.
IBarTypeMarkLine	Chart mark line.
IBarTypeMarkLineBlur	Fade style for the mark line; the fade rules follow those of the series.
IBarTypeMarkLineBlurLineStyle	The line style when the mark line fades.
IBarTypeMarkLineData	A single data point in the bar data array.
IBarTypeMarkLineEmphasis	Highlight style.



IBarTypeMarkLineLabel	The text of the mark line.
IBarTypeMarkLineLineStyle	Line style of the marker line.
IBarTypeMarkPoint	Chart annotation.
IBarTypeMarkPointBlur	The fade-out style for annotations follows the rules of the series to which they belong.
IBarTypeMarkPointData	The data array elements for the annotation.
IBarTypeMarkPointDataEmphasis	The highlighting style for this annotation data.
IBarTypeMarkPointDataLabel	The label style for the annotation data.
IBarTypeMarkPointEmphasis	The emphasis style for the annotation.
IBarTypeMarkPointItemStyle	The graphic style of the annotation.
IBarTypeSelect	Graphic and label styles when data is selected; effective when selectedMode is enabled.
IBarTypeSelectItemStyle	Graphic style when selected.
IBarTypeSelectLabelLine	Configuration of visual guides for the label when selected.
IBarType	A bar chart (also known as a column chart) is a common chart type that represents data magnitude through the height of the bars (or width, in the case of horizontal charts).
IBarTypeBackgroundStyle	The background style for each bar; this is only effective when showBackground is set to true.

5.1.17.3 IPieType

Pie charts are primarily used to show the proportion of data from different categories within a total. The arc of each segment represents the proportion of the data.

For scenarios where a chart contains multiple pie charts, you can use the Left, Right, Top, Bottom, Width, and Height properties to set the position and viewport size of each pie chart series.

Pie charts are better suited for representing relationships such as percentages relative to a total. If you only need to show the relative sizes of different data categories, it is recommended to use a bar chart. People are less sensitive to subtle differences in arc size than to subtle differences in length; therefore, you can configure the 'RoseType' property to display a Rose chart, using radius size to distinguish data magnitude.

The following is an overview of the functions of the PieType chart series sub-interfaces:

Interface Name	Function Overview
IPieTypeBlurLabelLine	Configuration of visual guide lines during fade-out.
IPieTypeData	Array of data content in the series.
IPieTypeDataBlur	Fade-out state for individual data items in a pie chart.
IPieTypeDataBlurItemStyle	The style of the graphic when a single data item in the pie chart fades out.
IPieTypeDataBlurLabelLine	Configuration of the visual guide lines for the label when a single data



	item fades out.
IPieTypeDataEmphasis	The highlighting style for individual data items.
IPieTypeDataEmphasisItemStyle	Graphic style for a single data point when highlighted in a pie chart.
IPieTypeDataEmphasisLabelLine	Configuration of the visual guide lines for the label when a single data item is highlighted.
IPieTypeDataItemStyle	The graphic style corresponding to a single data item.
IPieTypeDataLabel	The text label for a data item, used to provide explanatory information about the chart.
IPieTypeDataLabelLine	Configuration of the visual guide lines for the label associated with a single data item.
IPieTypeDataSelect	The selected state of an individual data item in a pie chart.
IPieTypeDataSelectItemStyle	The graphical style of a pie chart's individual data item when selected.
IPieTypeDataSelectLabelLine	Configuration of the visual guide lines for data labels when a single data item is selected.
IPieTypeDecal	The decal pattern for the chart.
IPieTypeDimensionsData	Use 'dimensions' to define information for each dimension of 'series.data' or 'dataset.source'.
IPieTypeEmphasis	Sector and label styles for pie chart highlights.
IPieTypeEmphasisItemStyle	The style of the chart when a sector is highlighted.
IPieTypeEmphasisLabel	Label when the chart is highlighted.
IPieTypeEmphasisLabelLine	Configuration of visual guides for the label when highlighted.
IPieTypeEmptyCircleStyle	Placeholder circle style.
IPieTypeItemStyle	Pie chart graphic style
IPieTypeLabel	Label on the pie chart segment, used to provide explanatory information about the data.
IPieTypeLabelLayout	Unified layout configuration for labels.
IPieTypeLabelLine	Configuration of visual guides for labels. Visual guides are displayed when the 'label-position' property is set to 'outside'.
IPieTypeMarkArea	Chart mark area, commonly used to highlight a specific range of data in a chart.
IPieTypeMarkAreaBlur	Fade-out style for the mark; the fade-out rules follow those of the series it belongs to.
IPieTypeMarkAreaData	The data array for the mark area.
IPieTypeMarkAreaDataBlur	The fade-out style for chart mark area data items.
IPieTypeMarkAreaDataEmphasis	Highlight style for chart mark area data items.
IPieTypeMarkAreaDataItemStyle	The style of this data item area; the itemStyles for the start and end items are combined.
IPieTypeMarkAreaDataLabel	Label style for the data items in the pie chart's mark area.
IPieTypeMarkAreaEmphasis	The highlighting style for the mark area.
IPieTypeMarkAreaItemStyle	The graphic style of the pie chart's mark area.
IPieTypeMarkAreaLabel	Configuration of the label text for the mark area.

IPieTypeMarkLine	Chart marker lines.
IPieTypeMarkLineBlur	Fade-out style for marks; the fade-out rules follow those of the series.
IPieTypeMarkLineData	Data for the mark line.
IPieTypeMarkLineDataBlur	The style of the marker data item during the fade-out.
IPieTypeMarkLineDataEmphasis	Highlight style for the marker data item.
IPieTypeMarkLineDataLabel	The style of the data item label; the labels for the start and end points are merged together.
IPieTypeMarkLineDataLineStyle	Line style for the mark line data item; the line styles for the start and end items are combined.
IPieTypeMarkLineEmphasis	The highlight style for the marker line.
IPieTypeMarkLineLabel	The label for the marker line.
IPieTypeMarkLineLineStyle	The line style of the chart marker line.
IPieTypeMarkPoint	Chart annotation.
IPieTypeMarkPointBlur	The fade-out style of the label; the fade-out rules follow those of the series it belongs to.
IPieTypeMarkPointData	Elements of the mark's data array.
IPieTypeMarkPointDataEmphasis	Highlight style for annotation data items.
IPieTypeMarkPointDataItemStyle	The marker style for the data points in the pie chart.
IPieTypeMarkPointDataLabel	Label for the data item
IPieTypeMarkPointEmphasis	The highlighting style for the data label.
IPieTypeMarkPointItemStyle	The graphic style of the pie chart's label.
IPieTypeMarkPointLabel	The label for the data point, which can be used to provide additional information about the data.
IPieTypeSelect	Selection state; effective when selectedMode is enabled.
IPieTypeSelectItemStyle	The ItemStyle when the pie chart is selected.
IPieTypeSelectLabel	Text label displayed when the chart fades out; can be used to provide explanatory information about the chart's data.
IPieTypeSelectLabelLine	Visual guide line when selected.
IPieType	Pie charts are primarily used to represent the proportion of different categories within a total.
IPieTypeBlur	The faded state of the pie chart.
IPieTypeBlurItemStyle	The graphic style of the pie chart when it is fading out.
IPieTypeBlurLabel	The label displayed when the chart fades out.

5.1.17.4 IRadarType

Radar charts are primarily used to visualize multivariate data, such as the analysis of various player attributes. Depends on the `IRadar` component. The following is a brief introduction to the functionality of the `RadarType` chart series sub-interfaces:

Interface Name	Function Overview
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IRadarTypeBlurItemStyle	Style settings for the fade-out effect of individual inflection point markers.
IRadarTypeBlurLabel	Fade-out label used to display data information about the graphic.
IRadarTypeData	Data for the radar chart.
IRadarTypeDataAreaStyle	The fill style for data item regions.
IRadarTypeDataBlur	Configuration for the fade-out state of a single inflection point.
IRadarTypeDataEmphasis	Highlighting state for a single data item.
IRadarTypeDataItemStyle	Style settings for a single inflection point marker.
IRadarTypeDataLabel	The label for a single data item, which can be used to provide additional information about the data in the graph.
IRadarTypeDataSelect	Selection status of a single data item in a radar chart.
IRadarTypeDecal	A decal pattern for the chart; this takes effect only when both `aria.enabled` and `aria.decal.show` are set to `true`.
IRadarTypeEmphasis	Highlight state; effective when `emphasis.focus` is enabled.
IRadarTypeEmphasisAreaStyle	The fill style for the highlighted area.
IRadarTypeEmphasisItemStyle	Graphic style settings for individual inflection point markers when highlighted.
IRadarTypeEmphasisLabel	Label displayed when highlighted, which can be used to provide additional data information about the graphic.
IRadarTypeItemStyle	Style of the line breakpoint marker.
IRadarTypeLabel	Labels on the graph, used to describe certain data information.
IRadarTypeLabelLayout	Unified layout configuration for labels.
IRadarTypeSelect	The selected state of the radar chart.
IRadarTypeSelectAreaStyle	Fill style for the selected area.
IRadarTypeSelectItemStyle	Style settings for an individual inflection point marker when selected.
IRadarTypeSelectLabel	Label displayed when selected, which can be used to provide additional information about the graphic.
IRadarType	Radar charts are primarily used to visualize multivariate data and rely on the radar component.
IRadarTypeAreaStyle	Area fill style.
IRadarTypeBlur	Configuration for the faded state; effective when emphasis.focus is enabled.
IRadarTypeBlurAreaStyle	Area fill style during the fade-out.

5.1.17.5 IGaugeType

Gauge charts, such as: car gauges, pressure gauges, temperature gauges, progress gauges, etc. The following is a brief introduction to the functions of the GaugeType graphical series sub-interfaces:

Interface Name	Function Overview
IGaugeTypeAxisLineLineStyle	Gauge axis line style.



IGaugeTypeAxisTick	Settings related to axis tick marks.
IGaugeTypeData	An array of data points in the series; each array element can be a single value.
IGaugeTypeDecal	Decal pattern for the chart; takes effect only when both `aria.enabled` and `aria.decal.show` are set to `true`, and requires that the `areaStyle` fill style be configured.
IGaugeTypeDetail	Gauge details, used to display data.
IGaugeTypeEmphasis	Highlighted gauge pointer style.
IGaugeTypeItemStyle	Gauge pointer style.
IGaugeTypeItemStyleEx	Gauge pointer style.
IGaugeTypeLabel	Text labels on the chart, which can be used to explain some of the chart's data.
IGaugeTypeLineStyle	Line style.
IGaugeTypeMarkArea	Mark area, commonly used to highlight a specific range of data in a chart, such as indicating the period during which an advertisement was aired.
IGaugeTypeMarkAreaBlur	Fade style for the mark area; the fade rules follow those of the series it belongs to.
IGaugeTypeMarkAreaData	The data array for the mark area.
IGaugeTypeMarkAreaEmphasis	Highlight style for the mark area; the highlighting rules follow those of the series.
IGaugeTypeMarkAreaLabel	Text configuration for the mark area, used to provide explanatory information about the chart data.
IGaugeTypeMarkLine	Chart mark line.
IGaugeTypeMarkLineBlur	Fade-out style for the mark line; the fade-out rules follow those of the series.
IGaugeTypeMarkLineData	Mark line data.
IGaugeTypeMarkLineDataBlur	Fade-out style for mark line data. The fade-out rules follow those of the series.
IGaugeTypeMarkLineDataEmphasis	The highlighting style for mark line data.
IGaugeTypeMarkLineDataLineStyle	The line style for the mark data; the lineStyle of the start and end points is combined.
IGaugeTypeMarkLineEmphasis	Highlight style for the mark line.
IGaugeTypeMarkPoint	Chart labels.
IGaugeTypeMarkPointBlur	Fade-out style for labels. The fade-out rules follow those of the series.
IGaugeTypeMarkPointData	The data array elements for the mark.
IGaugeTypeMarkPointEmphasis	The highlight style of the mark.
IGaugeTypePointer	Gauge pointer.
IGaugeTypeProgress	Displays current progress.
IGaugeTypeSplitLine	Split line style.



IGaugeTypeTitle	Dashboard title.
IGaugeType	Dashboard.
IGaugeTypeAnchor	Anchor point for the pointer on the dial.
IGaugeTypeAxisLabel	Scale label.
IGaugeTypeAxisLine	Configuration related to the gauge axis.

5.1.17.6 IScatterType

Scatter (Bubble) Chart. A scatter plot in a Cartesian coordinate system can be used to illustrate the relationship between the x and y values of data. If data items have multiple dimensions, values from other dimensions can be represented as bubbles using symbols of different sizes, or they can be distinguished by color. These features can be implemented in conjunction with the `IVisualMap` component.

Scatter (bubble) charts can be applied to Cartesian, polar, and geographic coordinate systems.

The following is a brief introduction to the functions of the ScatterType graphical series sub-interfaces:

Interface Name	Function Overview
IScatterTypeBlurLabelLine	Configuration of visual guide lines for labels during fading.
IScatterTypeData	Array of data content in the series.
IScatterTypeDataBlur	Fade-out style and label settings for individual data items.
IScatterTypeDataEmphasis	Highlighting state for bubble chart data items; effective when <code>emphasis.focus</code> is enabled.
IScatterTypeDataItemStyle	Style settings for individual data points (bubbles).
IScatterTypeDataLabel	Style settings for the text of a single data item.
IScatterTypeDataLabelLine	Configuration of visual guides for labels.
IScatterTypeDataSelect	Style configuration for a single data item when selected.
IScatterTypeDecal	The decal pattern for the graphic; this takes effect only when both <code>'aria.enabled'</code> and <code>'aria.decal.show'</code> are set to <code>'true'</code> . If set to <code>'none'</code> , no decal pattern is used.
IScatterTypeDimensionsData	Use <code>'dimensions'</code> to define the information for each dimension of <code>'series.data'</code> or <code>'dataset.source'</code> .
IScatterTypeEmphasis	The highlighting state of the bubble chart. Effective when <code>emphasis.focus</code> is enabled.
IScatterTypeEmphasisItemStyle	The graphic style when highlighted.
IScatterTypeEmphasisLabel	The label when highlighted.
IScatterTypeEmphasisLabelLine	Visual guide line configuration for the label when



	highlighted.
IScatterTypeItemStyle	Graph style.
IScatterTypeLabel	Labels on the scatter plot, which can be used to provide additional information about the data.
IScatterTypeLabelLayout	Unified layout configuration for labels.
IScatterTypeLabelLine	Configuration of visual guide lines for the label.
IScatterTypeMarkArea	Chart mark area, commonly used to mark a specific range of data in a chart.
IScatterTypeMarkAreaBlur	The fade-out style for annotations. The fade-out rules follow those of the series.
IScatterTypeMarkAreaBlurItemStyle	Style settings for individual data points (bubbles) when the marked area fades.
IScatterTypeMarkAreaBlurLabel	The style of labels on the graph when the marked area fades.
IScatterTypeMarkAreaData	Chart mark area data.
IScatterTypeMarkAreaDataBlur	The fading style for the mark area data.
IScatterTypeMarkAreaDataEmphasis	Highlight style for mark area data. Effective when emphasis.focus is enabled.
IScatterTypeMarkAreaDataItemStyle	Graphic (bubble) style settings for marked area data.
IScatterTypeMarkAreaDataLabel	Style of the mark area data item labels; the labels for the start and end items are merged together.
IScatterTypeMarkAreaEmphasis	Highlighting state for the mark area in the scatter plot.
IScatterTypeMarkAreaEmphasisItemStyle	The graphic style of the bubble chart mark area when highlighted.
IScatterTypeMarkAreaEmphasisLabel	The label displayed when the mark area is highlighted, which can be used to provide additional data information about the chart.
IScatterTypeMarkAreaItemStyle	The graphic style for this mark area.
IScatterTypeMarkAreaLabel	The style of the mark area label.
IScatterTypeMarkLine	Chart marker line.
IScatterTypeMarkLineBlur	Fade style for marks. The fade rules follow those of the series.
IScatterTypeMarkLineBlurLabel	Label style when the mark line fades.
IScatterTypeMarkLineBlurLineStyle	Line style when the marker fades.
IScatterTypeMarkLineData	Chart marker lines, commonly used to highlight a specific range of data in a chart.
IScatterTypeMarkLineDataBlur	The fade-out style for marker data. The fade-out rules follow those of the series.
IScatterTypeMarkLineDataEmphasis	The highlighting state of data points in a scatter plot.
IScatterTypeMarkLineDataLabel	The style of the data point label; the labels for the

	start and end points are merged together.
IScatterTypeMarkLineDataLineStyle	Line style for the mark line data
IScatterTypeMarkLineEmphasis	The highlighting style for the marker lines.
IScatterTypeMarkLineEmphasisLabel	Label style when the marker line is highlighted.
IScatterTypeMarkLineEmphasisLineStyle	The line style when the marker line is highlighted.
IScatterTypeMarkLineLabel	Label style for the marker line.
IScatterTypeMarkLineLineStyle	The line style of the marker line.
IScatterTypeMarkPoint	Chart annotation.
IScatterTypeMarkPointBlur	The blur style of the mark. The blur rules follow those of the series.
IScatterTypeMarkPointBlurItemStyle	The shape (bubble) style of the mark when it fades.
IScatterTypeMarkPointBlurLabel	Label style when the mark fades.
IScatterTypeMarkPointData	The data object for the annotation.
IScatterTypeMarkPointDataEmphasis	The highlight style for the annotation.
IScatterTypeMarkPointDataItemStyle	Graphic style of the annotation data.
IScatterTypeMarkPointDataLabel	Label style for the scatter data.
IScatterTypeMarkPointEmphasis	Highlight style for the mark.
IScatterTypeMarkPointEmphasisItemStyle	The graphic style when the mark is highlighted.
IScatterTypeMarkPointEmphasisLabel	The label style when the annotation is highlighted.
IScatterTypeMarkPointItemStyle	The graphic style of the annotation.
IScatterTypeMarkPointLabel	Label style for the mark.
IScatterTypeSelect	Selection state. Effective when selectedMode is enabled.
IScatterTypeSelectItemStyle	The graphic style of the bubble chart when selected.
IScatterTypeSelectLabel	Label style when a scatter plot is selected.
IScatterTypeSelectLabelLine	Configuration of visual guides for the label when the scatter chart is selected.
IScatterType	Scatter (Bubble) Chart.
IScatterTypeBlur	Fade style.
IScatterTypeBlurItemStyle	Graph style when fading.
IScatterTypeBlurLabel	Fade-out label style.

5.1.17.7 IMapType

Maps are primarily used for visualizing geographic data and are used in conjunction with the Map component to display data such as population density across different regions. Multiple series with the same map type will be displayed on the same map; in this case, the configuration settings of the first series are used as the rendering configuration for the map.

The following is an overview of the MapType graphical series sub-interface features:

Interface Name	Function Overview
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IMapTypeEmphasisEx	Polygon and label styles when highlighted.
IMapTypeItemStyle	Polygons in the map area Map area style.
IMapTypeLabel	Text labels on shapes, used to provide explanatory information about the shape's data.
IMapTypeLabelEx	Text labels on shapes, used to describe certain data information about the shapes.
IMapTypeLabelLayout	Unified layout configuration for labels.
IMapTypeLabelLine	Configuration of visual guides for labels.
IMapTypeLabelLineEx	Visual guide line configuration for labels.
IMapTypeMarkArea	Chart mark area, commonly used to highlight a specific range of data in a chart, such as indicating the period during which an advertisement was displayed.
IMapTypeMarkAreaData	Data array for the marked area.
IMapTypeMarkLine	Chart marker line.
IMapTypeMarkLineData	Data array for the marker line.
IMapTypeMarkPoint	Chart mark.
IMapTypeMarkPointData	Data array elements for map marks.
IMapTypeProjection	Custom map projection.
IMapTypeScaleLimit	Controls the limits of wheel zooming via the minimum and maximum zoom values; the default zoom level is 1.
IMapTypeSelect	Styles for polygons and labels when selected.
IMapTypeSelectEx	Polygon and label styles when selected.
IMapType	The map is primarily used for visualizing geographic data and works with the Map component to display data such as population density across different regions.
IMapTypeBlur	Polygon and label styles when faded.
IMapTypeData	Data content within the series.
IMapTypeEmphasis	Polygon and label styles in the highlighted state.

5.1.17.8 ISunburstType

Sunburst chart. Composed of multiple concentric rings, the inner ring serves as the parent node of the outer ring in the data structure. Therefore, it can represent both the proportion of parts to the whole, like a pie chart, and hierarchical relationships, like a tree diagram.

Data Drill-Down: Sunburst charts support data drill-down by default. This means that when a user clicks on a specific sector, that node becomes the root node for the display, and a circle appears in the center to return to the parent node. If you do not want the data drill-down feature, you can achieve this by setting ``series-sunburst.nodeClick`` to ``false``.

The following is an overview of the SunburstType chart series sub-interface features:



Interface Name	Function Description
ISunburstTypeDataEmphasis	Configuration of the state when data is highlighted; takes effect after `emphasis.focus` is enabled.
ISunburstTypeDecal	Decal pattern for the graphic; takes effect only when both `aria.enabled` and `aria.decal.show` are set to `true`. Requires the `areaStyle` fill style to be set to take effect.
ISunburstTypeEmphasis	Configuration for the highlighted state; takes effect when `emphasis.focus` is enabled.
ISunburstTypeItemStyle	Style for the sunburst-type segment.
ISunburstTypeItemStyleEx	Style of the sunburst-type segment.
ISunburstTypeLabel	A label on a graphic that can be used to provide explanatory information about the graphic's data.
ISunburstTypeLabelLayout	Unified layout configuration for labels.
ISunburstTypeLabelLine	Configuration of the label's visual guide lines.
ISunburstTypeLabelLineEx	Visual guide line configuration for labels.
ISunburstTypeLevel	Multi-level configuration.
ISunburstTypeLevelEmphasis	Highlight style for multi-level configuration.
ISunburstTypeSelect	This setting applies when selectedMode is enabled.
ISunburstType	A Sunburst chart consists of multiple concentric rings. In terms of data structure, the inner ring is the parent node of the outer ring. Therefore, it can represent both the proportion of parts relative to the whole, like a pie chart, and hierarchical relationships, like a tree diagram.
ISunburstTypeBlur	Fade-out state configuration; effective when `emphasis.focus` is enabled.
ISunburstTypeData	Data in the series